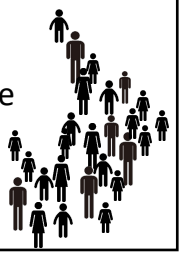
Demo·graphy

the origin in Greek

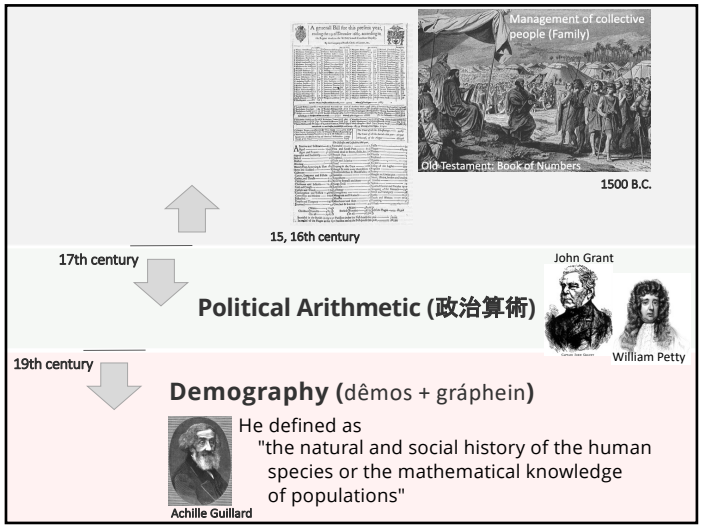
Demos + graphia

people (collectively) — writing, drawing, counting

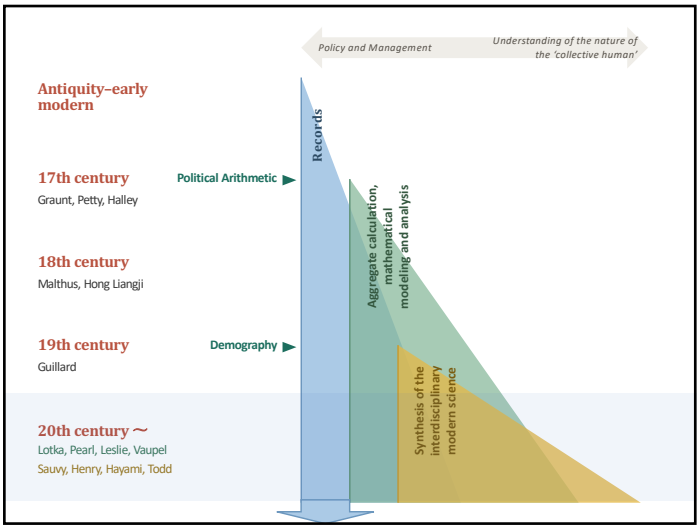
✓ Counting the number of people



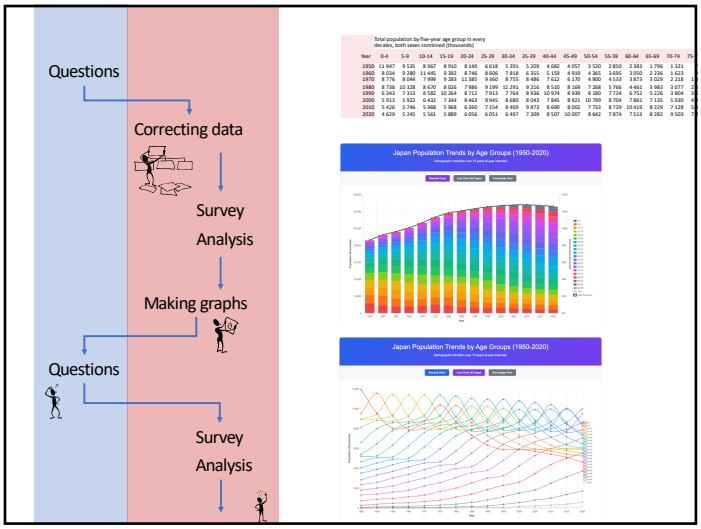
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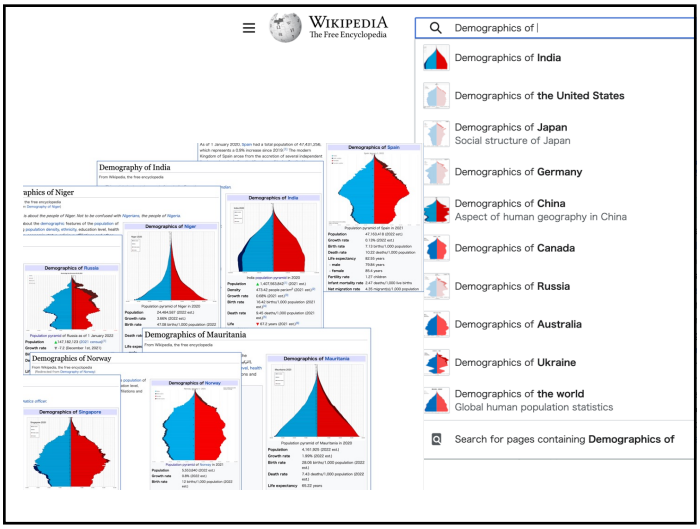
2



3



4



5

0. Introduction

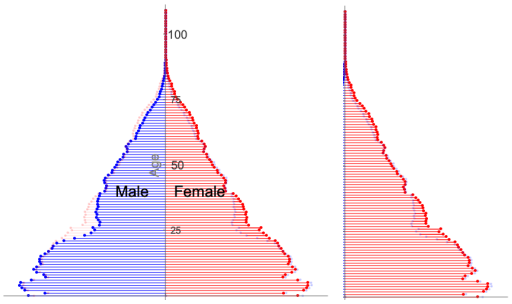
1. Nature of individual life: Death process
2. Nature of individual life: Childbearing

3. Characteristics of population that arise from the nature of individual's life

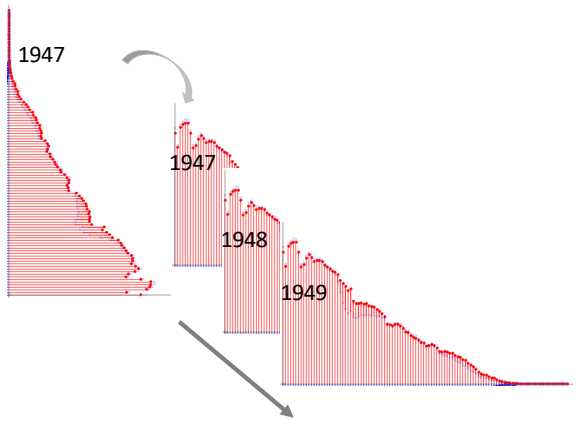
4. Traits of individuals defined in the population

5. Synthesis of the interdisciplinary science

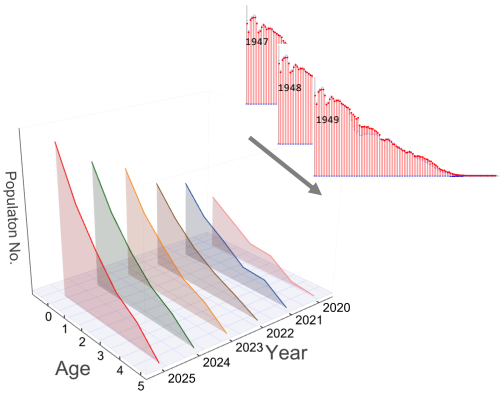
6



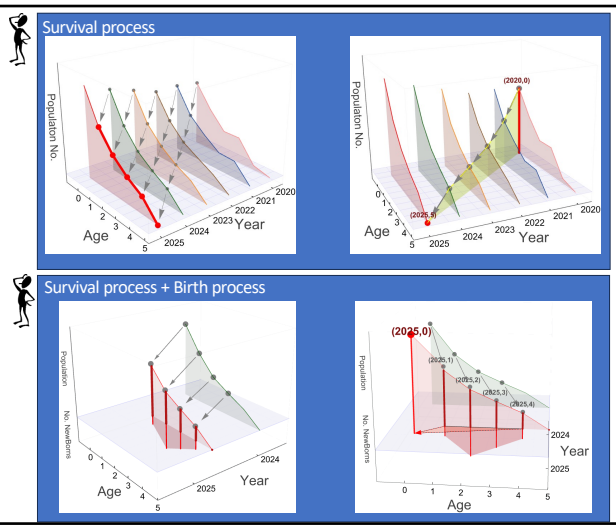
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8



9

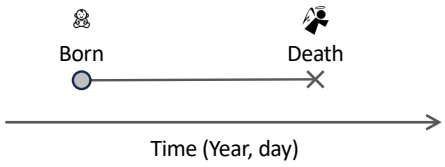


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0. Introduction
1. Nature of individual life: Death process
2. Nature of individual life: Childbearing
3. Characteristics of population that arise from the nature of individual's life
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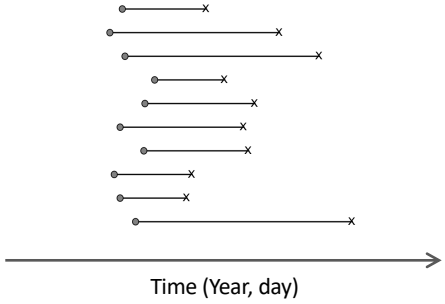
Life of an individual



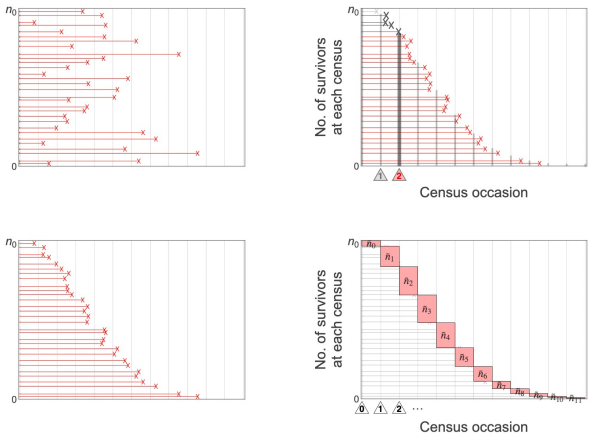
12



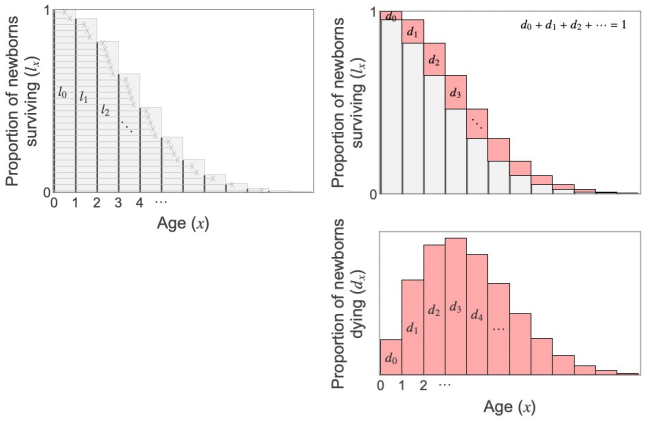
Life of people



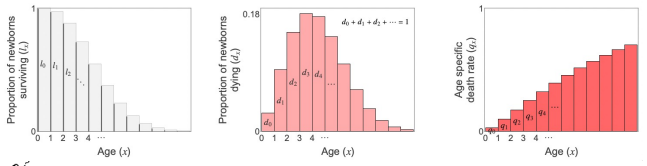
13



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How do you answer the questions?

- ✓ What are typical lifespans?
- ✓ To what extent do lifespans differ?

Quiz

Let's try to address this issue.

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Life expectancy:

Formula 1

$$= 1\,d_0 + 2\,d_1 + 3\,d_2 + \ldots$$

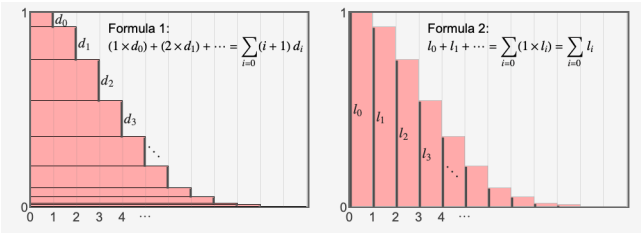
Formula 2

$$= l_0 + l_1 + l_2 + \cdots = \sum_{i=0} l_i$$

$$\begin{array}{l} 1\,d_0 \rightarrow \\ + \\ 2\,d_1 \rightarrow \\ + \\ 3\,d_2 \rightarrow \\ + \\ \vdots \end{array} \quad \begin{array}{l} d_0 \\ + \\ d_1 \\ + \\ d_2 \\ + \\ \vdots \end{array} \quad \begin{array}{l} d_1 \\ + \\ d_2 \\ + \\ \vdots \end{array} \quad \begin{array}{l} d_2 \\ + \\ \vdots \end{array} \quad \cdots$$

$$l_0 \quad + \quad l_1 \quad + \quad l_2 \quad + \quad \cdots$$

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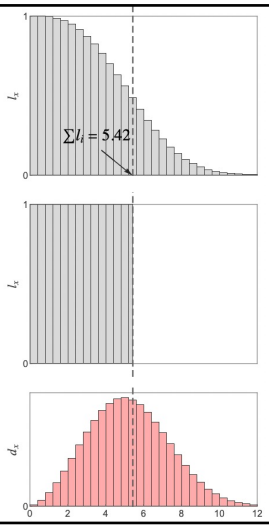


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Life expectancy

Formula 2

$$\sum_{i=0} l_i$$

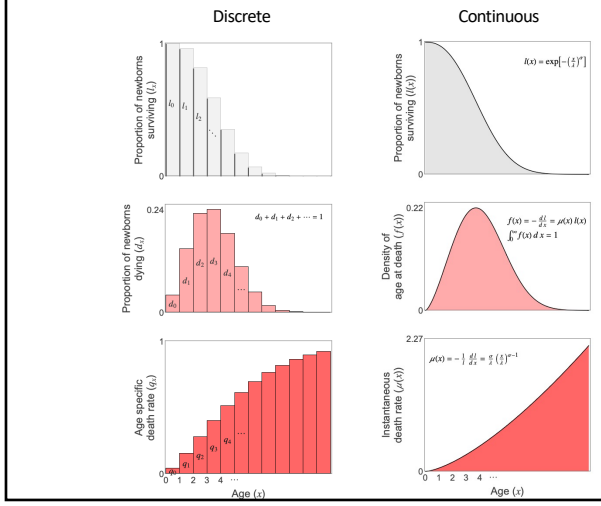


Formula 1

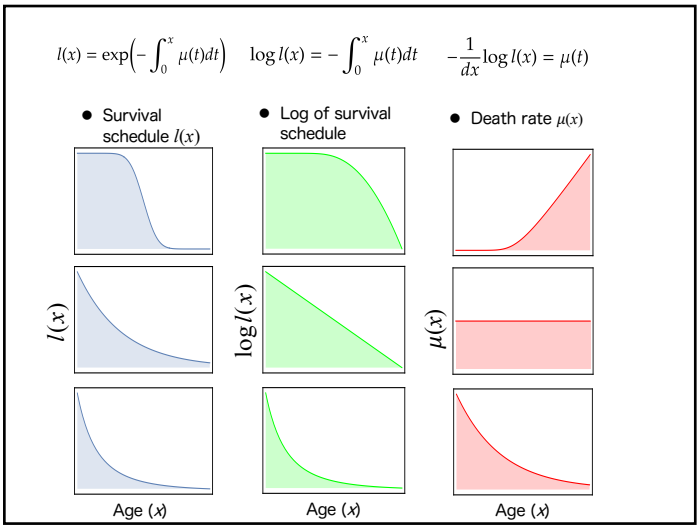
$$\sum_{i=0} = (i+1)\, d_i$$

weibull_life_expectancy_manipulate_7.wl

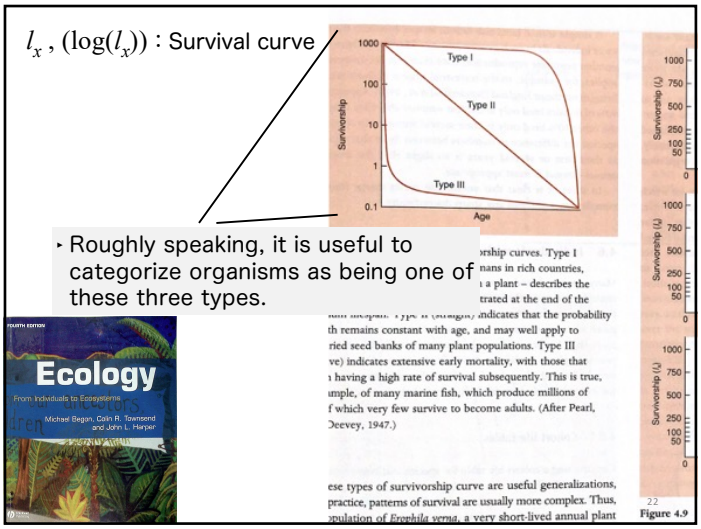
19



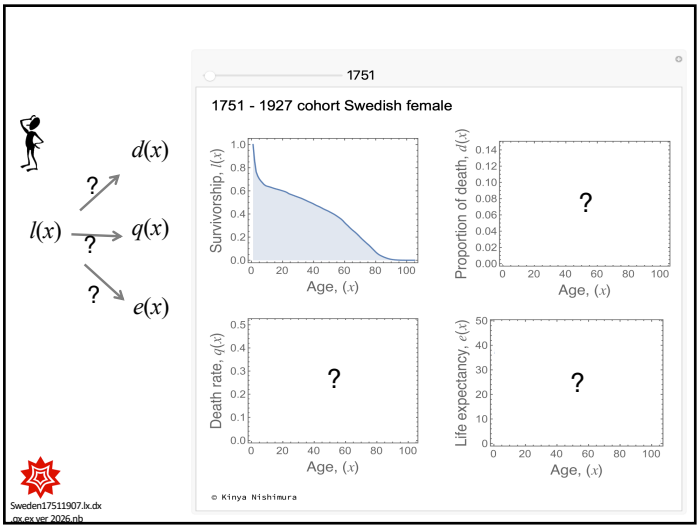
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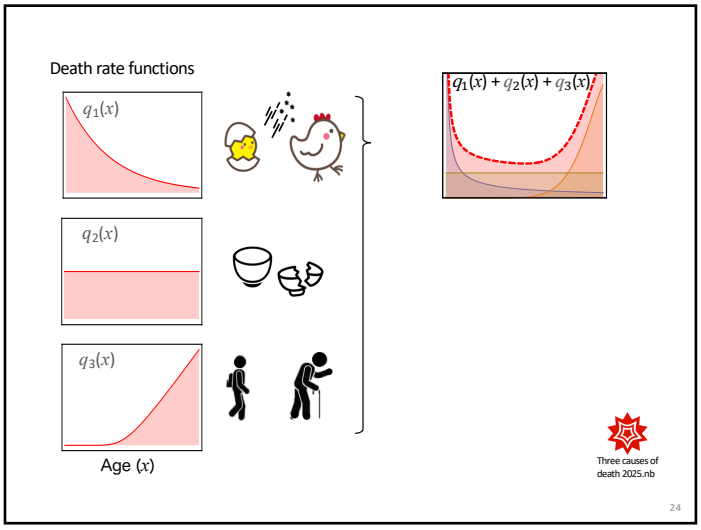
21



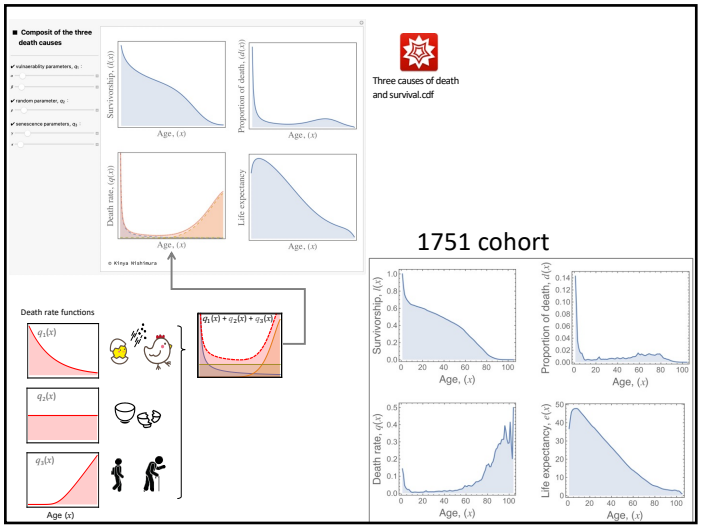
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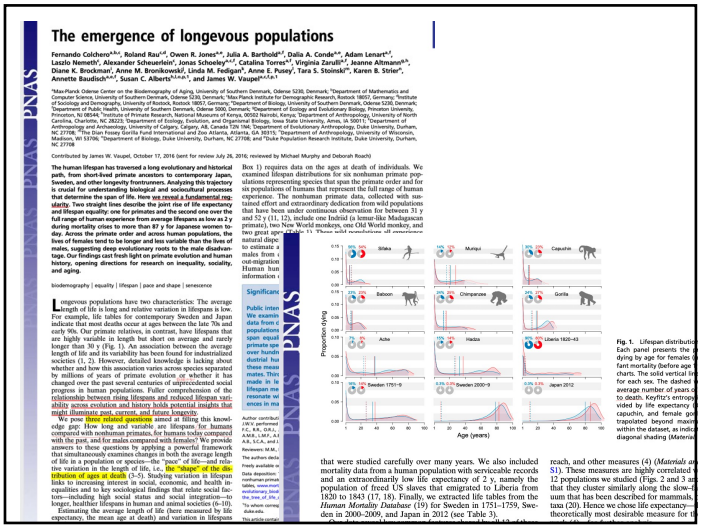
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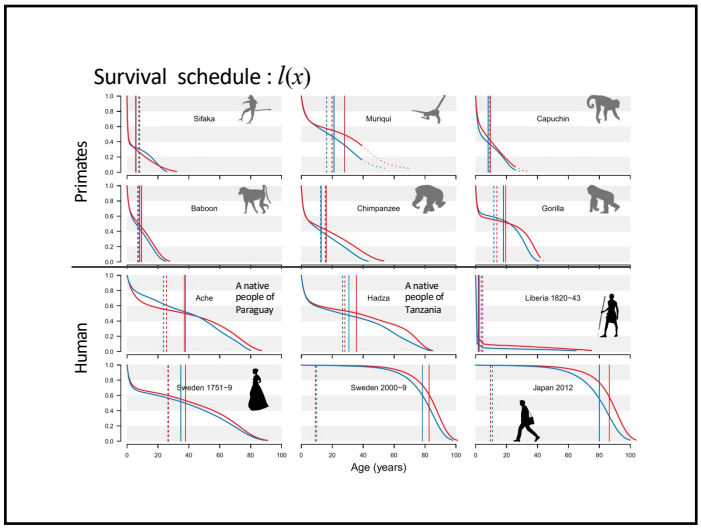
24



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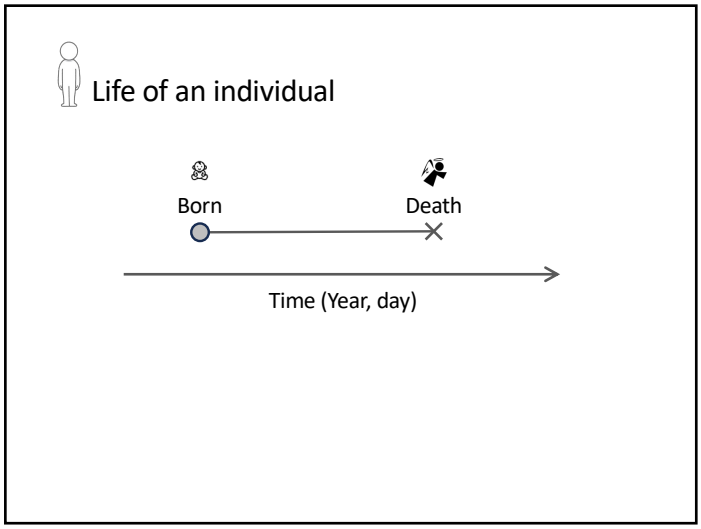
26



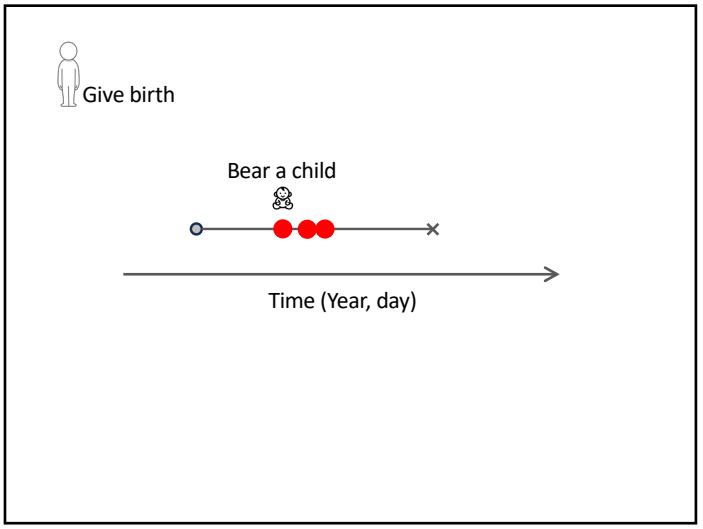
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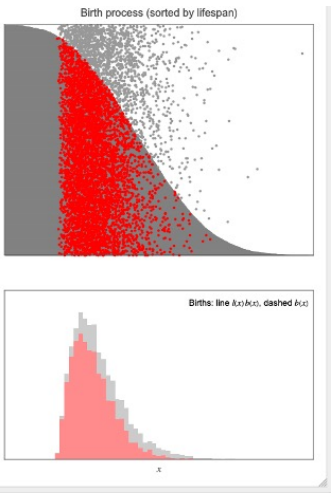


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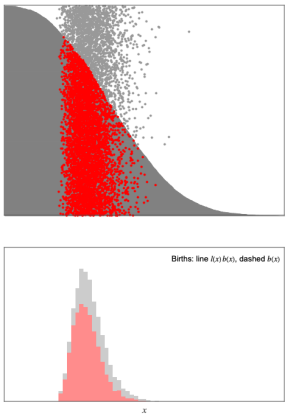


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Gray + Red $\int_0^\infty b(x)dx,\ \sum b_x$

Red $\int_0^\infty l(x)b(x)dx,\ \sum l_x b_x$

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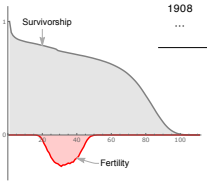
Summary

	Continuous	Discrete
Gross Reproduction	$\int_0^\infty b(x)dx$	$\sum_{x=0}^{\infty} b_x$
Net Reproduction	$\int_0^\infty l(x)b(x)dx$	$\sum_{x=0}^{\infty} l_x b_x$

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1891 cohort Life Table of Swedish females

Year	age (x)	n^x	Survivorship l_x	q^x	Fertility rate (No. of female birth) b_x
1891	0	100000	1.00000	0.09682	0.000000
1892	1	90318	0.90318	0.02954	0.000000
1893	2	87650	0.97046	0.01739	0.000000
1894	3	86126	0.98261	0.01292	0.000000
1895	4	85014	0.98709	0.00918	0.000000
...					
1904	13	81022	0.99663	0.00398	0.000020
1905	14	80699	0.99601	0.00411	0.000115
1906	15	80367	0.99589	0.00401	0.000710
1907	16	80045	0.99599	0.00502	0.003685
1908	17	79643	0.99498	0.00441	0.012755
...	...	...	...		...



Life expectancy = $\sum l_x = 57.78$

Expected lifetime reproduction = $\sum l_x b_x = 1.52$ (or total net fecundity)

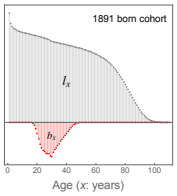
Total reproduction = $\sum b_x = 2.45$

Sweden1891Cohort.lx.bx.xlsx

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1891 cohort Life Table of Swedish females

Year	age (x)	n_x	l_x	d_x	q_x	b_x
1891	0	100000	1.00000	9682	0.09682	0.000000
1892	1	90318	0.90318	2668	0.02954	0.000000
1893	2	87650	0.97046	1524	0.01739	0.000000
1894	3	86126	0.98261	1113	0.01292	0.000000
1895	4	85014	0.98709	780	0.00918	0.000000
...						
1904	13	81022	0.99663	322	0.00398	0.000020
1905	14	80699	0.99601	332	0.00411	0.000115
1906	15	80367	0.99589	323	0.00401	0.000710
1907	16	80045	0.99599	402	0.00502	0.003685
1908	17	79643	0.99498	351	0.00441	0.012755
...						

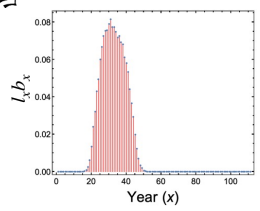


Quiz

- At what age does a woman give birth on average in her lifetime?
- What is the formula to get that age by using l_x and b_x ?

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- At what age does woman give birth on average in her lifetime? What is the formula to get the age by using l_x and b_x.



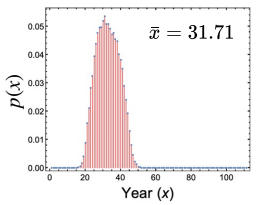
$$\sum l_x b_x = R_0$$

$$p(x) = \frac{l_x b_x}{\sum l_x b_x} = \frac{l_x b_x}{R_0}$$

$$\sum p(x) = 1$$

The average x of the probability distribution is

$$\sum xp(x)$$



1891 cohort Swedish female

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